# Functional Description of Diagnosis

## ZOTYE A01

## ABS Gen9

## Protocol: KWP2000 On CAN



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# 1 History

All corrections of misspelling are excluded of the following list.

| Ver. | Date | Editor | Chapter | Description of changes |
|---|---|---|---|---|
| 01.00 | 18.10.2010 | Sathish Muthu S | All | Initial Version<br>Based on China mainstream DIAF v1.1 |
| 01.01 | 15.03.2011 | Sathish Muthu S | 4<br>5.5.2<br>5.7.2<br><br><br><br>6<br><br><br><br><br>6.1.1<br>6.2<br><br>6.3.4<br>6.4.1<br><br>6.4.1 &<br>6.5.1<br>6.5.3<br>6.6.3<br><br><br>6.7.1<br><br><br>6.9.2<br><br>6.10.1<br>6.10.2<br>6.10.6<br><br>6.10.6 &<br>6.10.7<br>6.10.12.1<br>6.10.12.2 | - Abbreviation table is added.<br>- P2* time is updated to 5000ms and P3max time is updated to 10000ms.<br>- Stop Diagnostic session is removed and ECU Reset is added.<br>- P3max is updated as P3 CAN-ECU (MAX).<br>- P3 CAN-ECU (MAX) value is updated to 10 seconds.<br>- Service sections are updated.<br>- Service Stop Diagnostic session (SID 20h) is removed and ECU reset (SID 11h) is added.<br>- In service Start routine by LID, Periodic Actuator control (LID 02) is split in to Evac and Fill control (LID 02h) and Repair Bleed control (LID 03h).<br>- LIDs for Actuator test, Dynamic test and Wheel speed sensor test are updated.<br>- "StandardDiagnosticMode" is updated as "DefaultDiagnosticMode".<br>- ECU Reset service description is added. Stop Diagnostic session service description is removed.<br>- P2max is updated to P2 CAN-ECU.<br>- Description for "NumberOfFaultDetection" is modified.<br>- Comment for bit4 is updated.<br>- "DTCWarningLampCalibrationStatus" is changed to "DTCWarningLampStatus".<br>- In third remark byte2 is updated to byte1.<br>- Vehicle name and Customer part number is updated.<br>- BB number, version numberand supplier part number data type representation has been added.<br>- Byte(size) is modified as Size(Byte).<br>- Bit position is updated in Remark. Bit32 -> Bit31 and Bit31 -> Bit30.<br>- Remark 2 related to wheel and vehicle speed is added.<br>- Remark 1 is updated.<br>- In Remark 2, control status is updated to Control in progress(03h).<br>- Supported LID table is updated.<br>- Remark 1 system 8 is updated to system9.<br>- Description for Periodic actuator control is removed.<br>- Description for Evac and Fill and Repair Bleed is added.<br>- Control status 03h and 05h are added.<br>- Bit position is updated in Remark. Bit32 -> Bit31 and Bit31 -> Bit30. Supported data table |



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| | | | | |
|---|---|---|---|---|
| | | | | is updated. |
| | | | 6.10.12.3 | - LID 03h is added. |
| | | | 6.10.12.5 & 6.10.12.6 | - Remark related to the byte 3...10 is added. |
| | | | 6.10.12.7 | - Value of speed returned when WSS faulure detected is updated from FFFFh to 7FFFh. |
| | | | 6.7.4 & 6.11.4 | - Remark for NRC 78h is added. |
| | | | 7.1 | - DTC 0x501008 - ECU error (general software problem) is added. |
| | | | 8 | - Diagnostic flow chart is updated with supported services. |
| | | | | The above changes are done based on the latest China MS Gen9 KWP on CAN DIAF v1.8 |



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## 2.1 International and National Documents

The general requirements of CAN diagnosis communication protocol are based on the Keyword Protocol 2000 Design Standard requirements, specified in the following documents:

1. ISO 14229 : Road Vehicles – Diagnostic Systems: Diagnostic service specification
2. ISO 14230-1: Road Vehicles – Diagnostic Systems: KWP2000 – Physical layer
3. ISO 14230-2: Road Vehicles – Diagnostic Systems: KWP2000 – Data Link Layer
4. ISO 14230-3: Road Vehicles – Diagnostic Systems: KWP2000 - Part 3 Implementation
5. ISO 15765-2: Road Vehicles – Diagnostics on CAN: Network Layer Services
6. ISO 15765-3: Road Vehicles – Diagnostics on CAN: Application Layer Services

Remark:
Requirements in the above referenced documents may not be fully implemented. The document Functional Description of Diagnosis is the agreed specification for the implemented customer diagnosis.



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# 3 Introduction

CAN diagnosis provides the capability of quickly accessing the information and functions of the control unit over the **C**ontroller **A**rea **N**etwork (CAN), without disconnecting the cable harness. CAN diagnosis provides services to:

- Identify the ECU
- Read input signals of the ECU
- Write output signals to the actuators
- Read out ECU-internal information, such as the contents of the fault memory

CAN diagnosis can be used for end-of-line testing or in the servicing workshop. The testing time can be reduced substantially by using the services provided by CAN diagnosis.

This document describes the diagnosis functions and specifications of the Normal, EOL and project specific Diagnostic Session (Application Software).

# 4 Abbreviations

| Abbreviations | Description |
|---|---|
| ABS | Antilock Braking Systems |
| AV | Outlet Valve |
| BLS | Brake Light Switch |
| DTC | Diagnostic Trouble Code |
| EBD | Electronic Brake force Distribution |
| ECU | Electronic Control Unit |
| EV | Inlet Valve |
| IGN | Ignition |
| LSB | Least Significant Bit |
| MSB | Most Significant Bit |
| PID | Parameter Identifier |



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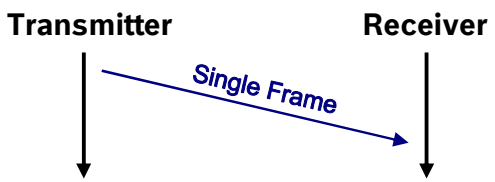
# 5 Communication on CAN

## 5.1 Data Stream

The ECU supports CAN diagnostic messages using the Unacknowledged Segmented Data Transfer (USDT) mechanism as specified in ISO 15765-2. Accordingly the following two types of CAN data streams are supported:

### 5.1.1 Non-segmented data stream

Consists of a Single Frame (SF) of non-segmented CAN message. This can be used for messages with maximum data length of 7 data bytes




### 5.1.2 Segmented data stream

A segmented data stream consists of multiple frames of segmented CAN messages. A segmented data stream must be used if the maximum data length exceeds 7 data bytes in case of normal addressing.

The data stream comprises a First Frame (FF) and a number of Consecutive Frames (CF) containing the subsequent data bytes, following the First Frame. The receiver has to send a Flow Control (FC) after reception of the First Frame.

The Flow Control contains information regarding Block Size and the Separation Time. The Block Size (BS) is the maximum number of Consecutive Frames the receiver allows the sender to send, before waiting for an authorisation (Additional Flow Control Frames) from the receiver to continue transmission. The Separation Time (Stmin) is the (minimum) time the sender has to wait between the transmissions of two consecutive frames.



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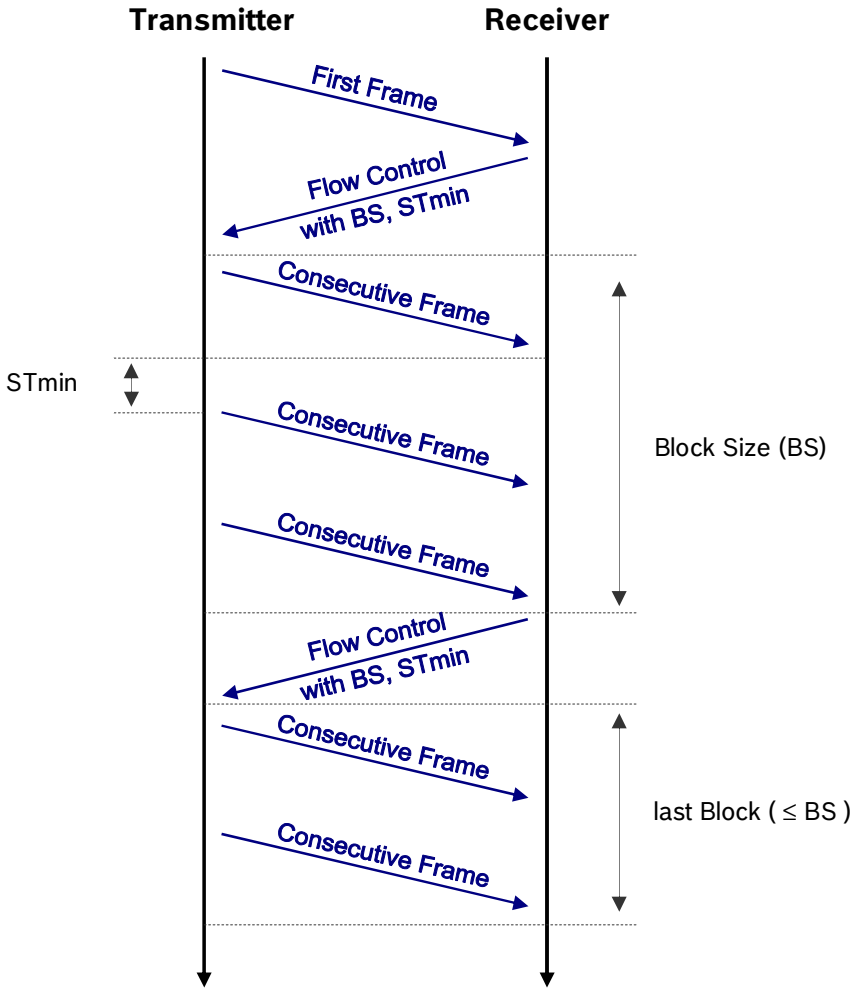




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## 5.2 Message Types

Each Single Frame (SF), First Frame (FF), Consecutive Frame (CF) or Flow Control (FC) has an 8-byte Data Field (DLC of CAN message is set to 8) consisting of the following parts:

- Protocol Control Information (PCI)
  indicates the frame type, number of frames, data length etc.
- Transmission data (Data)
  contains data to be sent

| Data Field Structure | | | | | |
|---|---|---|---|---|---|
| **Frame Type** | **Byte 1** | | **Byte 2** | **Byte 3** | **Byte 4 – 8** |
| | **Bits 7-4** | **Bits 3-0** | | | |
| Single Frame (SF) | 0000 | SF_DL | DB 1 | DB 2 | DB 3 – 7 |
| First Frame (FF) | 0001 | FF_DL | | DB 1 | DB 2 – 6 |
| Consecutive Frame (CF) | 0010 | SN | DB $x_1$ | DB $x_2$ | DB $x_3$-$x_7$ |
| Flow Control (FC) | 0011 | FS | BS | Stmin | N/A |

*SF_DL: 4 Bit Single Frame Data length*
*SN: Sequence Number*
*BS: Block Size*
*DB: Data byte*
*FF_DL: 12 Bit First Frame Data length*
*FS: Status of data stream*
*Stmin: Minimum Separation Time*

### 5.2.1 Single Frame

Single Frame message type is used for data blocks with a maximum data length of 7 Bytes and is identified by the upper nibble of PCI set to 0000b. Single Frame Data length SF_DL is computed as:

**Single Frame Data Length (SF_DL):**

| Hex Value | Description |
|---|---|
| 0h – 7h | Single Frame Data length (SF_DL) contains number of data Bytes |

### 5.2.2 First Frame

First Frame message type is used for data blocks more than 7 Bytes long, in combination with one or more Consecutive Frames and is identified by the upper nibble of PCI set to 0001B. First Frame Data length FF_DL is computed as:

**First Frame Data Length (FF_DL):**

| Hex Value | Description of FF_DL |
|---|---|
| 000h – 007h | Reserved |
| 008h – FFFh | First Frame Data length (FF_DL) contains number of data bytes. 12 Bit FF_DL theoretically supports sending of up to 4096 data bytes. However the current implementation does not need to support this maximum of 4096 data bytes. |



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### 5.2.3 Consecutive Frame

Consecutive Frames follow First Frame and is identified by the upper nibble of PCI set to 0010B. The lower nibble of the PCI has a Sequence Number (SN) which is determined as follows:

- First Frame is regarded as containing Sequence Number 0.
- Hence the first Consecutive Frame always holds Sequence Number 1.
- Sequence Number is incremented by one for every additional Consecutive Frame.
- A Flow Control frame sent "in between" does not affect the Sequence Number counting.
- When the Sequence Number reaches the value of 15, the following Consecutive Frame will be sent with Sequence Number set to 0.

**Definition of Sequence Number (SN):**

| **Protocol Data name** | FF | CF | CF | CF | CF | CF | CF | CF | CF |
|---|---|---|---|---|---|---|---|---|---|
| **SN (hex)** | 0h | 1h | 2h | ... | Eh | Fh | 0h | 1h | ... |

| Hex Value | Description |
|---|---|
| 0h – Fh | Sequence Number (SN)<br>The Sequence Number has a range of 0 to 15. SN-Value is stored to the lower 4 bits of the Byte 1. |

### 5.2.4 Flow Control

After the First Frame or the last Consecutive Frame of a block of frames when further CFs are required to complete the data transfer, a Flow Control message must be sent by the receiver node, and is identified by the upper nibble of PCI set to 0011b. With the Flow Control, the receiving node notifies the transmitting node how many messages it can receive between two Flow Control frames (maximum Block Size, Bsmax) and at what rate between Consecutive Frames (minimum Separation Time, Stmin). The lower nibble of the Flow Control PCI shall have a Flow Control Status (FS) set to "Continue To Send (CTS)".

**Stream Status (FS):**

| Hex Value | Description |
|---|---|
| 0h | Continue To Send (CTS)<br>The Receiver is ready and awaits the BS announced maximal amount of Consecutive Frames |
| 1h | Wait (WT)<br>The FlowControl Wait parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to "1". It shall cause the sender to continue to wait for a new FlowControl N_PDU and to restart its N_BS timer. |
| 2h | Overflow (OVFLW)<br>The FlowControl Overflow parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to "2". It shall cause the sender to abort the transmission of a segmented message and make an N_USData.confirm service call with the parameter <N_Result>=N_BUFFER_OVFLW. This N_PCI FlowStatus parameter value is only allowed to be transmitted in the FlowControl N_PDU that follows the FirstFrame N_PDU and shall only be used in case the message length FF_DL of the received FirstFrame N_PDU |



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| | |
|---|---|
| | exceeds the buffer size of the receiving entity. |

**Block Size (Bsmax):**

| Hex Value | Description |
|---|---|
| 00h | Block Size (Bsmax)<br>No more Flow Controls will be sent during the segmented Message. All Consecutive Frames will be sending without a stop for an additional Flow Control to the Receiver. |
| 01h – FFh | Block Size (Bsmax)<br>Defines the maximal amount of Consecutive Frames till next Flow Control. |

**Separation Time (Stmin):**

| Hex Value | Description |
|---|---|
| 00h – FFh | Separation Time (Stmin)<br>The Separation Time defines the minimal allowed time (in ms) between 2 Consecutive Frames. |

## 5.3 Addressing Modes

### 5.3.1 Physically addressing Mode

All Physically addressed messages will be received by the ECU on ECU_PHY_ADDR (see 5.5.1 section below for supported CAN ID). It is possible to execute a service supported in that active diagnostic session of the ECU, when the request is made using this mode. Response from the ECU is sent on TESTER_ADDR.

## 5.4 Diagnostic Sessions

The ECU supports the following Diagnostic Sessions:

- **Normal / Default Session**
  At ignition ON, the ECU enables the Normal diagnostic session by default. In this session both the controller functions and diagnostic services are active. But this session does not permit the execution of all supported diagnostic services of the EOL and specific Diagnostic Session (see “Availability of Services” section below for services supported during this session).

- **EOL Diagnostic Session**
  The Diagnostic session for assembly line.



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## 5.5 Overview

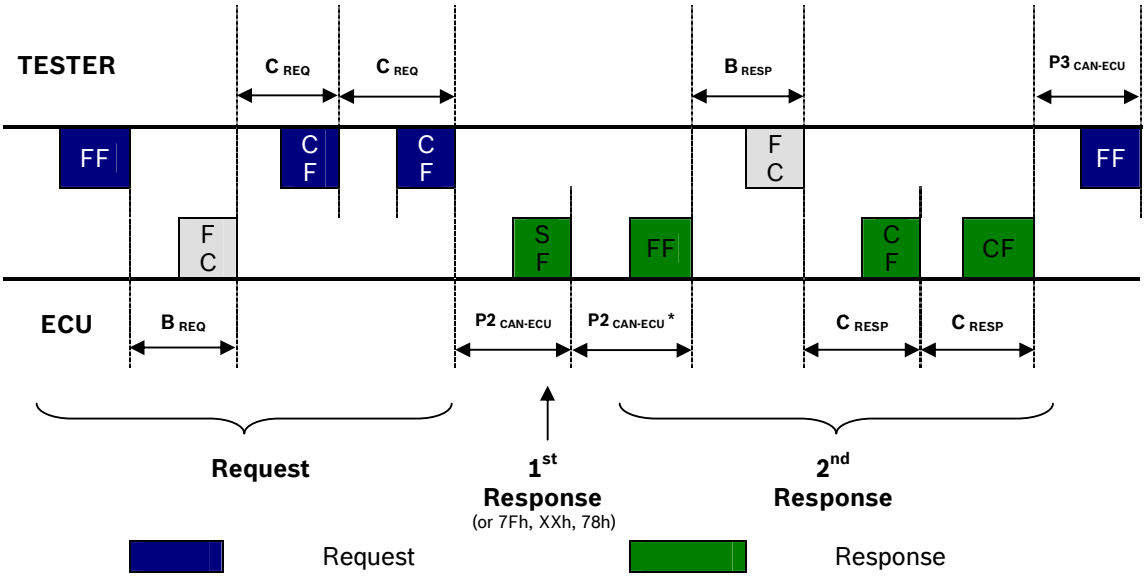
The CAN communication between Tester and ECU is only possible in one direction at a time. In a Half-Duplex diagnostic session the Tester is the master and the ECU is the slave. The master will send requests and the slave – if addressed – will answer with a response.

### 5.5.1 Communication Addresses

| Addressing Type | Name | Hex Value<br>ZOTYE A01 |
|---|---|---|
| Tester CAN receive message ID for Diagnosis | TESTER_ADDR | 712h |
| ECU CAN receive message ID for Physical Addressing | ECU_PHY_ADDR | 702h |

### 5.5.2 Communication Timing Parameters

Network Layer Timing:



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Parameters for Physical Layer, Data Link Layer and Network Layer:

| Parameter | Name | Value | |
|---|---|---|---|
| Communication speed (baud rate) | BR-CAN | 500 kBd ± 2 kBd | |
| Max Timeout to access the network<br>(time for transmission of the CAN frame on the sender side) | As | 25 ± 2 ms | |
| Blocksize – the maximum number of Consecutive Frames the receiver allows the sender before waiting for a Flow Control message | Bsmax | 0 | |
| Separation time – The Receiver can set the Stmin value as 10ms. The Transmitter shall wait for more than Receiver Stmin. Value before sending the next Consecutive Frame | Stmin | 10ms | |
| **Tester Request** | | Min | Max |
| Time between reception of First Frame from Tester and transmission of Flow Control from ECU.<br>The maximum timing is defined under the condition that the CAN Bus load is able to process the ECU transmission. | $B_{REQ}$ (Br) | NA | 50 ms# |
| Time between transmission of Flow Control and reception of next Consecutive Frame or Time between reception of two Consecutive Frames | $C_{REQ}$ (Cs) | | 50 ms# |
| Time between end of reception of a Tester Request and start of an ECU-Response | $P2_{CAN\text{-}ECU}$ | 0 ms | 50 ms |
| Time between end of transmission of an ECU response with Negative Response Code 78h (Response Pending) and start of next ECU response | $P2_{CAN\text{-}ECU}$* | 0 ms | 5000 ms |
| **ECU Response** | | Min | Max |
| Time between transmission of First Frame and reception of Flow Control from Tester | $B_{RESP}$ (Bs) | NA | 100 ms |
| Time between reception of Flow Control and transmission of following Consecutive Frame or Time between transmission of two Consecutive Frames | $C_{RESP}$ (Cr) | 0 | 100 ms |
| Time between end of an ECU-Response and start of a Tester-Request in Normal Diagnostic Session | $P3_{CAN\text{-}ECU\text{-}StdDiag}$ | NA | Infinite |
| Time between end of an ECU-Response and start of a Tester-Request in Extended Diagnostic Session | $P3_{CAN\text{-}ECU\text{-}ExtDiag}$ | NA | 10000 ms |

#The maximum timing is defined under the condition that the CAN Busload is able to process the ECU transmission.

Note:
When the tester sends a request and the tester does not expect any response (e.g. functionally addressed request or a physical addressed message with “No response required” option), the tester has to wait

$$P3_{CAN\text{-}ECU\text{-}func} = P2_{CAN\text{-}ECU\ (Max)} + P3_{CAN\text{-}ECU\ (Min)} + P_{Service_Execution_Time}$$

before sending the next request.



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## 5.6 Communication Failures

### 5.6.1 Unexpected Messages

As a general rule, the arrival of an unexpected message or a message with illegal format from any node will be ignored.

### 5.6.2 Missing Messages

If the ECU senses a communication failure (e.g. timing violations of the Network Layer) during a tester request, the ECU discards the request and no response will be sent.

The tester has to monitor that no ECU-response has been sent within **P2** $_{\text{CAN-ECU}}$, and hence the tester can repeat his previous request.

During Extended Diagnostic Session, if no request is received from the tester for **P3** $_{\text{CAN-ECU (MAX)}}$ (criteria for broken communication wire) ECU will exit to Normal Session.

## 5.7 Condition and Behaviour of the ECU

### 5.7.1 Speed Limit

The ECU monitors a speed limit during the entire EOL diagnosis mode. Speed limit for the tester address (see 5.5.1) is 10 Km/h.

### 5.7.2 Exit Conditions

The ECU will terminate an active EOL diagnostic mode in the following cases:

- The speed limit is reached. Note that speed limit can be disabled by using the InputOutputControlByLocalIdentifier service (see 6.9).
- An ECU Reset request is received (See 6.2).
- The ECU does not receive any tester message for more than 10 seconds (**P3** $_{\text{CAN-ECU (MAX)}}$ Timeout)

EOL diagnostic session is started using the StartDiagnosticSession service. When an EOL diagnostic session is aborted, all EOL diagnostic actuations (eg: lamps, Valves) will be switched off.



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# 6 Diagnostic Service List

ECU and Tester should support the diagnostic services listed below.

| Diagnostic Service Name | Req. SID | LID | Std. Diag. Mode | ECU EOL Mode | |
|---|---|---|---|---|---|
| | | | | With Security Locked | With Security Unlocked |
| StartDiagnosticSession | 10h | | ◆ | ◆ | ◆ |
| ECU Reset | 11h | | ◆ | ◆ | ◆ |
| ClearDiagnosticInformation | 14h | | ◆ | ◆ | ◆ |
| ReadStatusOfDTC | 17h | | ◆ | ◆ | ◆ |
| ReadDTCByStatus | 18h | | ◆ | ◆ | ◆ |
| ReadECUID | 1Ah | | ◆ | ◆ | ◆ |
| ReadDataByLocalID | 21h | | | | |
| Supported PID | | 00h | ◆ | ◆ | ◆ |
| Filling-in status of the Hydraulic block | | 02h | ◆ | ◆ | ◆ |
| Status of Test of End-Of-Line | | 03h | ◆ | ◆ | ◆ |
| Wheel speed FL | | 04h | ◆ | ◆ | ◆ |
| Wheel speed FR | | 05h | ◆ | ◆ | ◆ |
| Wheel speed RL | | 06h | ◆ | ◆ | ◆ |
| Wheel speed RR | | 07h | ◆ | ◆ | ◆ |
| Vehicle speed | | 08h | ◆ | ◆ | ◆ |
| Valve status | | 09h | ◆ | ◆ | ◆ |
| Pump Motor status | | 10h | ◆ | ◆ | ◆ |
| BLS Status | | 11h | ◆ | ◆ | ◆ |
| Battery Voltage | | 1Ch | ◆ | ◆ | ◆ |
| SecurityAccess | 27h | | | ◆ | ◆ |
| InputOutputControlByLocalIdentifier | 30h | | | ◆ | ◆ |
| StartRoutineByLocalID | 31h | | | | |
| Supported LID | | 00h | | ◆ | ◆ |
| Delayed Actuator Control | | 01h | | ◆ | ◆ |
| Evac And Fill Control | | 02h | | ◆ | ◆ |
| Repair Bleed Control | | 03h | | ◆ | ◆ |
| Actuator Test | | 04h | | ◆ | ◆ |
| Dynamic Test | | 05h | | ◆ | ◆ |
| Test of the wheel speed sensors | | 06h | | ◆ | ◆ |
| WriteDataByLocalID | 3Bh | | | | |
| Filling Status of the Hydraulic block | | 02h | | | ◆ |
| Status of Test of End-Of-Line | | 03h | | | ◆ |
| TesterPresent | 3Eh | | ◆ | ◆ | ◆ |

Note: Sending request for unsupported services will return “7F XX 11” as negative response.
XX – Service ID sent in the request.



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## 6.1 StartDiagnosticSession

### 6.1.1 Purpose

The service StartDiagnosticSession is used to select a specific ECU DiagnosticMode (called session). Depending on the session selected, actions are authorized, prohibited or limited. The following sessions are defined:

| PARAMETER: DiagnosticMode | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | DefaultDiagnosticMode | 81h |
| 1 | ECUEOLMode | 83h |

**Remark:** In the Default Session the ABS functionality is fully available.

**Remark:** In the EOL mode the ABS functionality is not available. Switching to EOL mode is only possible when the ECU is not in regulation.

### 6.1.2 Request

| TESTER → ECU: [SERVICE] - Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 10h |
| 1 | DiagnosticMode | xxh |

### 6.1.3 Positive Response

| ECU → TESTER: [SERVICE] – Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 50h |
| 1 | Diagnostic Mode | xxh |

### 6.1.4 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 10h |
| 2 | **Response Code**<br>Sub Function Not Supported / Invalid Format<br>ConditionsNotCorrect/RequestSeqError<br>BusyRepeatRequest | <br>12h<br>22h<br>21h |



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## 6.2 ECUReset

### 6.2.1 Purpose

The service ECUReset is used to restart the ECU.

| PARAMETER: ResetMode | | |
|---|---|---|
| Size (Byte) | Description | Value |
| 1 | PowerOn | 01h |

### 6.2.2 Request

| TESTER → ECU: [SERVICE] - Request | | |
|---|---|---|
| Byte | Contents | Value |
| 0 | Service Identifier | 11h |
| 1 | ResetMode | 01h |

### 6.2.3 Positive Response

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| Byte | Contents | Value |
| 0 | Service Positive Response Identifier | 51h |
| 1 | ResetMode | 01h |

### 6.2.4 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| Byte | Contents | Value |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 11h |
| 2 | **Response Code**<br>SubFunctionNotSupported/InvalidFormat<br>RequestCorrectlyReceived/ResponsePending | <br>12h<br>78h |



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## 6.3 ClearDiagnosticInformation

### 6.3.1 Purpose

The service ClearDiagnosticInformation is used to clear the fault memory. The parameter GroupOfDTC is used to indicate which ECU has to clear its fault memory.

### 6.3.2 Request

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 14h |
| 1 | GroupOfDTC – High byte | FFh |
| 2 | GroupOfDTC –Low byte | 00h |

### 6.3.3 Positive Response

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 54 |
| 1 | GroupOfDTC – High byte | FFh |
| 2 | GroupOfDTC –Low byte | 00h |

### 6.3.4 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 14h |
| 2 | **Response Code**<br>SubFunctionNotSupported/InvalidFormat<br>BusyRepeatRequest<br>RequestCorrectlyReceived/ResponsePending | <br>12h<br>21h<br>78h |

**Remark:** If the erasing of the EEPROM needs more than $P2_{CAN\text{-}ECU}$, the ECU will send a negative response 78h (RequestCorrectlyReceived-ResponsePending) until it has been finished.



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## 6.4 ReadStatusOfDTC

### 6.4.1 Purpose

The service ReadStatusOfDTC is used to read the status of a specific DTC from the failure memory.

| PARAMETER: StatusOfDTC | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | StatusOfDTC | xxh |

| StatusOfDTC | |
|---|---|
| **Position (Bit)** | **Description** |
| 0..3 | • DTCFaultSymptom<br>• 0000b = “No fault symptom available for this DTC”<br>• 0001b = “Above maximum threshold”<br>• 0010b = “Below minimum threshold”<br>• 0100b = “No signal”<br>• 1000b = “Invalid signal” |
| 4 | • 0b = Test completed / not complete for this DTC bit not supported. Set to 0b. |
| 5..6 | • DTCStorageState – Storage state of the error path:<br>• 00b = “No DTC detected at time of request”<br>• 01b = “Error path is stored in the FCM but currently not defective”<br>• 10b = “DTC maturing – intermittent at time of request”<br>• 11b = “Error path is stored in the FCM and currently defective” |
| 7 | • DTCWarningLampStatus, indicates whether a warning lamp is controlled by the path entered<br>• 0b = “Error path is not controlling a warning lamp”<br>• 1b = “Error path is controlling a warning lamp” |

| PARAMETER: NumberOfFaultDetections | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | If a fault remains current in each new ignition cycle, NumberOfFaultDetections gets increased by one until the fault is not deleted. (max value is FFh) | xxh |

**Remark:** The fault path is indicated by the stored DTC. When saving a new fault, a check is first conducted in order to establish whether the same DTC has already been saved at any point in the fault memory. If this is the case, the DTC is saved at the same position. Otherwise, it is saved at the next free position in the fault memory.



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### 6.4.2 Request

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 17h |
| 1 | DTC – High Byte | xxh |
| 2 | DTC – Middle Byte | xxh |
| 3 | DTC – Low Byte | xxh |

### 6.4.3 Positive Response

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 57h |
| 1 | NumberOfDTC | 01h |
| 2 | DTC – High Byte | xxh |
| 3 | DTC – Middle Byte | xxh |
| 4 | DTC – Low Byte | xxh |
| 5 | StatusOfDTC | xxh |
| 6 | NumberOfFaultDetections | xxh |
| 7 | Vehicle Speed (1LSB = 1km/h, FFh = Invalid) | xxh |
| 8 | ECU power supply (1LSB = 80mV, FFh = Invalid) | xxh |

### 6.4.4 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 17h |
| 2 | **Response Code**<br>Sub Function Not Supported / Invalid Format<br>BusyRepeatRequest<br>RequestCorrectlyReceived/ResponsePending | <br>12h<br>21h<br>78h |

**Remark:** If there is no saved fault corresponding to the requested DTC, the ECU is sending a negative response with the parameter 12h.



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## 6.5 ReadDTCByStatus

### 6.5.1 Purpose

The service ReadDTCByStatus is used to read all stored DTCs from the failure memory.

| StatusOfDTC | |
|---|---|
| **Position (Bit)** | **Description** |
| 0..3 | DTCFaultSymptom<br>• 0000b = “No fault symptom available for this DTC”<br>• 0001b = “Above maximum threshold”<br>• 0010b = “Below minimum threshold”<br>• 0100b = “No signal”<br>• 1000b = “Invalid signal” |
| 4 | • 0b = Test completed / not complete for this DTC bit not supported. Set to 0b. |
| 5..6 | DTCStorageState – Storage state of the error path:<br>• 00b = “No DTC detected at time of request”<br>• 01b = “Error path is stored in the FCM but currently not defective”<br>• 10b = “DTC maturing – intermittent at time of request”<br>• 11b = “Error path is stored in the FCM and currently defective” |
| 7 | DTCWarningLampStatus, indicates whether a warning lamp is controlled by the path entered<br>• 0b = “Error path is not controlling a warning lamp”<br>• 1b = “Error path is controlling a warning lamp” |

**Remark:** The entire fault memory of the ECU can accommodate up to 10 different errors simultaneously.

**Remark:** Old DTCs will be deleted after 200 ignition cycle. The Ignition cycle of the fault and current ignition cycle will be compared. If the difference is greater than or equal to 200, then the fault is deleted from the EEPROM.

### 6.5.2 Request

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 18h |
| 1 | StatusOfDTC | 00h |
| 2 | GroupOfDTC – High Byte | FFh |
| 3 | GroupOfDTC – Low Byte | 00h |



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### 6.5.3 Positive Response

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 58h |
| 1 | NumberOfDTC | xxh |
| 2..4 | DTC | xxxxxxh |
| 5 | StatusOfDTC | xxh |
| … | DTC | xxxxxxh |
| … | StatusOfDTC | xxh |

**Remark:** According to the NumberOfDTC more than one DTC can be sent.

**Remark:** For every DTC 4 bytes are sent.

**Remark:** If no DTC is saved the response ends at byte 1 with NumberOfDTC at the value 00h.

### 6.5.4 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Content** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 18h |
| 2 | **Response Code**<br>Sub Function Not Supported / Invalid Format<br>BusyRepeatRequest | <br>12h<br>21h |



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## 6.6 ReadECUIdentification

### 6.6.1 Purpose

The service ReadECUIdentification is used to read identification information from the ECU.
The parameter IdentificationOption is used to indicate which information has to be read.

| PARAMETER: IdentificationOption | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | ReportECUIdentificationTable | 80h |

### 6.6.2 Request

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 1Ah |
| 1 | IdentificationOption | 80h |

### 6.6.3 Positive Response

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 5Ah |
| 1 | Local Identifier | 80h |
| 2..4 | Vehicle Name (ASCII, 3 bytes) → A01 : 0x41 0x30 0x31 | xxxxxxh |
| 5..7 | System Name (ASCII, 3 bytes) → ABS : 0x41 0x42 0x53 | xxxxxxh |
| 8..10 | BBNumber (BCD, 3 Bytes) → BB90543: 0x09 0x05 0x43 | xxxxxxh |
| 11..14 | S/W Version (BCD, 4 bytes) → V01.02.01.01 : 0x01 0x02 0x01 0x01 | xxxxxxxxh |
| 15..19 | Supplier Specific Part Number (BCD, 5 bytes)<br>→ 0265805057 : 0x02 0x65 0x80 0x50 0x57 | xx…xxh |
| 20..32 | Customer Specific Part Number (ASCII ) 13 bytes<br>3550100-A01 : 0x33 0x35 0x35 0x30 0x31 0x30 0x30 0x2D 0x41 0x30 0x31 0x20 0x20 | xx…xxh |

**Remark:** Byte 15... 32 will be read from EEPROM.
**Remark:** If the customer part number length is less than the allocated bytes, then the remaining bytes will be updated with the value 0x20(ASCII value for SPACE).

### 6.6.4 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Content** | **Value** |
| 0 | Negative response Identifier | 7Fh |
| 1 | Request Service Identifier | 1Ah |
| 2 | **Response Code**<br>General Reject<br>SubFunctionNotSupported/InvalidFormat<br>ResponsePending | <br>10h<br>12h<br>78h |



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**Remark:** If the reading of the EEPROM needs more than $P2_{CAN\text{-}ECU}$, the ECU will send a negative response 78h (RequestCorrectlyReceived-ResponsePending) until it has finished.



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## 6.7 ReadDataByLocalIdentifier

### 6.7.1 Purpose

This service allows the tester to request current data from the ECU using specific data IDs.

| PARAMETER: RecordLocalidentifier | | | |
|---|---|---|---|
| **Request Data byte (Size)** | **Description** | **Response Databyte (Size)** | **Value (Record LocalIdentifier)** |
| **1** | Supported PID for 01 – 20<br>ABS → 7F 81 80 10 | 4 | 00h |
| **1** | Filling-in status of the hydraulic block<br>Filling-in status:<br>• 00h : Filling-in not completed<br>• AAh : Filling-in completed and ok<br>• EEh : Filling-in completed and not ok | 1 | 02h |
| **1** | Test of End-Of-Line<br>End of Line Test status:<br>• 00h : End of Line Test not completed<br>• AAh : End of Line Test completed and ok<br>• EEh : End of Line Test completed and not ok | 1 | 03h |
| **1** | Wheel Speed FL<br>Range 0…255 km/h 1LSB = 0.0563km/h | 2 | 04h |
| **1** | Wheel Speed FR<br>Range 0…255 km/h 1LSB = 0.0563 km/h | 2 | 05h |
| **1** | Wheel Speed RL<br>Range 0…255 km/h 1LSB = 0.0563 km/h | 2 | 06h |
| **1** | Wheel Speed RR<br>Range 0…255 km/h 1LSB = 0.0563 km/h | 2 | 07h |
| **1** | Vehicle Speed<br>Range 0…255 km/h 1LSB = 0.0563 km/h | 2 | 08h |
| **1** | Valve Status<br>Byte 1: Bit 0: EVFL - Inlet valve front left - 1h ON / 0h OFF<br>Bit 1: AVFL - Outlet valve front left - 1h ON / 0h OFF<br>Bit 2: EVFR - Inlet valve front right - 1h ON / 0h OFF<br>Bit 3: AVFR - Outlet valve front right - 1h ON / 0h OFF<br>Bit 4: EVRL - Inlet valve rear left - 1h ON / 0h OFF<br>Bit 5: AVRL - Outlet valve rear left - 1h ON / 0h OFF<br>Bit 6: EVRR - Inlet valve rear right - 1h ON / 0h OFF<br>Bit 7: AVRR - Outlet valve rear right - 1h ON / 0h OFF | 1 | 09h |
| **1** | Pump Motor Status<br>FFh ON / 00h OFF | 1 | 10h |
| **1** | BLS Status<br>FFh ON / 00h OFF | 1 | 11h |
| **1** | Battery Voltage<br>Range 8…19.89V 1LSB = 0.080 V/bit | 1 | 1Ch |



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**Remark:** If a valve relay is down then the Valve status (PID 09h) value is not valid.

**Remark:** If WSS fault is logged then the Wheel Speed (PID 04h~07h) or Vehicle Speed (PID 08h) value is not valid. (7FFFh is returned)

**Remark:** Supported PID (00h) gives the information about the supported local identifier from 01h – 20h (Bit 31 - PID 01h, Bit 30 – PID 02h ... Bit 0 – PID 20h).

For Example
- Request 21 00
- Response 61 00 7F 81 80 10 (ABS)

### 6.7.2 Request

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 21h |
| 1 | RecordLocalIdentifier | xxh |

### 6.7.3 Positive Response

| ECU → TESTER: [SERVICE] – Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 61h |
| 1 | RecordLocalIdentifier | xxh |
| 2 | Data record #1 Byte 1 (MSB) | xxh |
| … | … | … |
| 2 + r | Data record #1 Byte r (LSB) | xxh |

### 6.7.4 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Content** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 21h |
| 2 | **Response Code**<br>General Reject<br>Sub Function Not Supported / Invalid Format<br>Response Pending | <br>10h<br>12h<br>78h |

**Remark:** If the reading of the EEPROM needs more than $P2_{CAN\text{-}ECU}$, the ECU will send a negative response 78h (RequestCorrectlyReceived-ResponsePending) until it has finished.



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## 6.8 Security Access

### 6.8.1 Purpose

The service SecurityAccess is used to get access to secured services by changing to different level of security. Changing the level of security is a two step procedure:

1. Tester requests seed, ECU sends seed
2. Tester sends the key, ECU sends a positive response (meaning the key was valid) and changes to the new security level

**Remark:** After two false attempts to change the security level the ECU is locked for 10 seconds. During these 10 seconds period the ECU will give negative response with NRC22h for any request to unlock the ECU. After 10 seconds ECU will allow 2 more false attempts before the lock of next 10 seconds.

| PARAMETER: AccessModeRequestSeed | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | Request Seed for General Security Level 1 | 01h |

| PARAMETER: AccessModeSendKey | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | Send Key for General Security Level 1 | 02h |

| PARAMETER: SecurityAccessStatus | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | SecurityAccessAllowed | 34h |

### 6.8.2 Request Step 1

| TESTER → ECU: [SERVICE] - Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 27h |
| 1 | AccessModeRequestSeed | 01h |

### 6.8.3 Positive Response Step 1

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 67h |
| 1 | AccessModeRequestSeed | 01h |
| 2 | Seed Byte 1 – High Byte | xxh |
| 3 | Seed Byte 2 | xxh |
| 4 | Seed Byte 3 | xxh |
| 5 | Seed Byte 4 – Low Byte | xxh |



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### 6.8.4 Request Step 2

| TESTER → ECU: [SERVICE] - Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 27h |
| 1 | AccessModeSendKey | 02h |
| 2 | Key Byte 1 – High Byte | xxh |
| 3 | Key Byte 2 | xxh |
| 4 | Key Byte 3 | xxh |
| 5 | Key Byte 4 – Low Byte | xxh |

### 6.8.5 Positive Response Step 2

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 67h |
| 1 | AccessModeSendKey | 02h |
| 2 | SecurityAccessStatus | 34h |

### 6.8.6 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Content** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 27h |
| 2 | **Response Code**<br>Sub Function Not Supported / Invalid Format<br>Conditions Not Correct / Request Sequence Error<br>Wrong key | <br>12h<br>22h<br>35h |



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## 6.9 InputOutputControlByLocalIdentifier

This request is used to actuate a single control. The parameter ControlOption is used to indicate whether the control has to be temporary, permanent or whether it has to be stopped. When using ControlOption "Start temporary control" the control runs for 2s. When using ControlOption "Start permanent control" the control runs until it is stopped by sending the same request with ControlOption set to "Stop control".

Supported LIDs for Single Output Control (InputOutputLocalIdentifier)

| PARAMETER: InputOutputControlByLocalIdentifier | | | | |
|---|---|---|---|---|
| **Size (Byte)** | **Description** | **Value** | **Temp.** | **Perm.** |
| 1 | Single control - Motor (Pump) | 01h | Yes | No |
| 1 | Single control - Front left valve (inlet) | 02h | Yes | No |
| 1 | Single control - Front right valve (inlet) | 03h | Yes | No |
| 1 | Single control - Rear left valve (inlet) | 04h | Yes | No |
| 1 | Single control - Rear right valve (inlet) | 05h | Yes | No |
| 1 | Single control - Front left valve (outlet) | 06h | Yes | No |
| 1 | Single control - Front right valve (outlet) | 07h | Yes | No |
| 1 | Single control - Rear left valve (outlet) | 08h | Yes | No |
| 1 | Single control - Rear right valve (outlet) | 09h | Yes | No |
| 1 | Single control - Speed limit disable (for $V_{Veh} \geq 10$km/h) | 1Eh | No | Yes |
| 1 | Single control - Disable valve relay shutdown | 1Fh | No | Yes |

### 6.9.1 Request

| TESTER → ECU: [SERVICE] - Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | ServiceIdentifier | 30h |
| 1 | InputOutputLocalIdentifier | xxh |
| 2 | **InputOutputControlParameter**<br>Start temporary control 01h<br>Stop control 02h<br>Status interrogation 03h<br>Start permanent control 04h | xxh |

Two ways of control can be distinguished:

Temporary: The actuator will be switched on for 2 s.
Permanent: The actuator will be switched on. It can be switched off (stopped) by sending a "Emergency Stop Request" (command 02H).



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### 6.9.2 Positive Response

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 70h |
| 1 | InputOutputLocalIdentifier | xxh |
| 2 | **Status**<br>Started 01h<br>Control completed 02h<br>Control In Progress 03h<br>Stopped 04h<br>Failure Or Not Started 05h | xxh |

**Remarks**

- If the tester is trying to start a **new control while another actuation is still being executed**, the ECU responds with the Positive response – Failure Or Not started (05h).This is not applicable for temporary control while Permanent control is in progress and vice versa. If the tester is trying to start a **new control while the same actuation is still being executed**, the ECU responds with the Positive response- Control in Progress(03h).
- **LIDs $1E and $1F should be used with care**, these LIDs will disable parts of the safety software. Therefore, the ECU may be damaged or the vehicle may not be safe to drive (e.g. if the speed limit during diagnostics is disabled and then the valves are activated while driving the car).

**If the valves are activated for long time**, A valve overheat protection functionality is implemented. When Valves are actuated for more than a specified time then actuation will be disabled .When Valves cool down actuation will be activated again. This is done through up count and down count mechanism. Overheat protection for valves avoid failures in the field.

**Remark:** After a start permanent control request in combination with Input Output Control By Local Identifier values 1Eh and 1Fh the ECU will respond to a status interrogation request in combination with Input Output By Local Identifier values 1Eh and 1Fh with a positive response including the ControlStatus 03h (control In Progress).

### 6.9.3 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 30h |
| 2 | **Response Code**<br>Sub Function Not Supported / Invalid Format<br>Conditions Not Correct / Request Sequence Error | <br>12h<br>22h |



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## 6.10 StartRoutineByLocalIdentifier

### 6.10.1 Purpose

The service StartRoutineByLocalIdentifier is used to control actuators connected to the ECU. Depending on the parameter StartRoutineLocalIdentifier a specific function is started (01h to 06h). The following values for the parameter StartRoutineLocalIdentifier are defined:

| Size (Byte) | Description | Value |
|---|---|---|
| 1 | GetSupportedLID | 00h |
| 1 | DelayedActuatorControl | 01h |
| 1 | Evac&Fill Routine | 02h |
| 1 | RepairBleed Control | 03h |
| 1 | ActuatorTest | 04h |
| 1 | DynamicTest | 05h |
| 1 | Test of the wheel speed sensors | 06h |

| PARAMETER: ControlOption | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | Start Routine | 01h |
| 1 | Stop Routine | 02h |
| 1 | Request routine results | 03h |

### 6.10.2 Requests

**Remark:** If the valves are activated for too long the ECU can be damaged. The maximum activation time of the valves must not exceed 30 seconds at an ambient temperature of 30°C. Note that Bosch has changed the overheat protection strategy for system 9 (outlet valves no longer toggling after 1s). Using this service repeatedly can damage the valves or pump. Please keep this information in mind when using this service.

**Remark:** If the ControlOption is set to “Stop control” the request must only have 3 bytes: Service Identifier, StartRoutineLocalIdentifier, ControlOption.

**Remark:** For control actions with an executing time exceeding $P2_{CAN\text{-}ECU}$ the ECU normally responds positive with ControlStatus = Control in progress within $P2_{CAN\text{-}ECU}$. During executing time the tester has to ask for the control result by using the ControlOption = Status interrogation. The ECU will either send a response with the ControlStatus = Control in progress or with the ControlStatus = Control complete and in addition if applicable with a result. The service requests TesterPresent and EmergencyStop will as well be positively responded.



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### 6.10.3 Emergency Stop

At any time it is possible to interrupt the control of the output which is being tested.

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 31h |
| 1 | StartRoutineLocalIdentifier | xxh |
| 2 | ControlOption = Stop Control | 02h |

### 6.10.4 StartRoutineLocalIdentifier = GetSupportedLID

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 31h |
| 1 | StartRoutineLocalIdentifier | 00h |



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### 6.10.5 StartRoutineLocalIdentifier = DelayedActuatorControl

This request is used to control multiple drivers simultaneously. For each control two states are defined: state 1 and state 2. When the request was received the selected controls are set in state 1. After the activation time they are set to state 2.

In comparison to the Actuator Test this service allows the tester to actuate one or more valves for a specified time. This feature allows the tester to run the customer specified End Of Line routine sequence instead of Pre-defined Periodic Actuator Control sequence.

| PARAMETER: ValveStates | |
|---|---|
| **Position (Bit)** | **Description** |
| 0 | Inlet front left |
| 1 | Outlet front left |
| 2 | Inlet front right |
| 3 | Outlet front right |
| 4 | Inlet rear left |
| 5 | Outlet rear left |
| 6 | Inlet rear right |
| 7 | Outlet rear right |

| PARAMETER: DriverStates | |
|---|---|
| **Position (Bit)** | **Description** |
| 0 | Unused |
| 1 | Unused |
| 2 | Unused |
| 3 | Unused |
| 4 | Unused |
| 5 | Unused |
| 6 | Pump |
| 7 | Unused |

**Remark:** For each control the corresponding bit has to be set to 1 to switch the actuator on and to 0 to switch the actuator off.



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| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 31h |
| 1 | StartRoutineLocalIdentifier | 01h |
| 2 | ControlOption | 01h |
| 3 | ValveStates (State 1) | xxh |
| 4 | DriverStates (State 1) | xxh |
| 5 | Activation time (1LSB = 20ms) | xxh |
| 6 | ValveStates (State 2) | xxh |
| 7 | DriverStates (State 2) | xxh |

**Remark:** A "Failure" response has one of the following reasons: At least one valve or one driver doesn't exist or the control is not possible at this time because of safety reasons.

**Remark:** A control that has been started by a delay actuator control request can be stopped by a request with the ControlOption set to "Stop control" (Emergency Stop, 31h 01h 02h).

**Remark:** At each end of this service the tester has to send the following request to turn off all controls: 31h 01h 01h 00h 00h 00h 00h 00h.

### 6.10.6 StartRoutineLocalIdentifier = EvacAndFill

This request is used to fill and drain the hydraulic unit.

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 31h |
| 1 | StartRoutineLocalIdentifier | 02h |
| 2 | ControlOption | 01h |
| 3 | Evac and fill time | xxh |
| 4 | Time until start of pump motor | xxh |

**Remark:** Evac &Fill time has to be between 2 s and 150 s. If the requested time > 150 seconds, the ECU uses the maximum allowed value(150 s). If the requsted time<2seconds,the Routine will not run. Only pump run will happen based on the byte4(parameter 'Time until start of pump motor'). The duration time selection only supports multiple of 2 (as one EvacAndFill cycle takes 2 seconds). Odd numbers are cut down to the next lower even value (i.e. if the routine is requested to run 15 s, it will actually complete in 14 s).After evac and filling time, the pump will run for 2 additional seconds,if the delay time in byte4 is not FFh.If total Evac&Fill time <' Time until start of pump motor' ,then the ECU responds Negatively with NRC 12h.



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### 6.10.7 StartRoutineLocalIdentifier = RepairBleeding

This request is used to fill and drain the hydraulic unit.

| TESTER → ECU: [SERVICE] – Request | | | |
|---|---|---|---|
| **Byte** | **Contents** | **Value** | **Comment** |
| 0 | Service Identifier | 31h | |
| 1 | StartRoutineLocalIdentifier | 03h | |
| 2 | ControlOption | 01h | |
| 3 | Repair bleeding phase (1..4) | xxh | Phase 1..4 |
| 4 | Repetition counter(3..5) | xxh | No. of loops: 3..5 |

**Remark:**
Repair bleeding phase
The repair bleeding function should always be executed in increasing phase sequence starting with Phase 1. A repetition of the same phase is allowed, but no phase shall be skipped. If this is not assured, the pump might be damaged due to dry run.
Repetition counter
This parameter defines the number of inner loop repetitions in phase 3and 4. This parameter is always required, but only valid for Phase 3 and 4, for other phases, it is ignored.



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### 6.10.8 Parameter descriptions for Actuator Test and Dynamic Test

Following parameter descriptions are valid for the ActuatorTest and for the Dynamic Test

| PARAMETER: ResultSelection | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | Without result | 00h |
| 1 | With result | FFh |

| PARAMETER: ActuationValue | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 2 | This parameter specifies the amount of actuation. Resolution: 1 bit = 1% | xxxxh |

| PARAMETER: ActuatorNumber | | |
|---|---|---|
| Size (Byte) | **Description** | **Value** |
| 2 | No actuation* | 0000h |
| 2 | Inlet valve front left | 0030h |
| 2 | Outlet valve front left | 0032h |
| 2 | Inlet valve front right | 0034h |
| 2 | Outlet valve front right | 0036h |
| 2 | Inlet valve rear right | 0038h |
| 2 | Outlet valve rear right | 003Ah |
| 2 | Inlet valve rear left | 003Ch |
| 2 | Outlet valve rear left | 003Eh |
| 2 | Pump motor relay | 0022h |
| 2 | ABS Safety Lamp | 0002h |
| 2 | EBD Safety Lamp | 000Eh |

**Remark:** * Use if less than four actuators are to be actuated by an actuator test or dynamic test request.

| PARAMETER: ActuationTime | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 2 | This parameter specifies the time between the first and the second actuation. Resolution: 1 bit = 10 ms | Xxxxh |

| PARAMETER: WaitTime | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 2 | This parameter specifies the waiting time. Resolution: 1 bit = 10 ms | Xxxxh |



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### 6.10.9 StartRoutineLocalIdentifier = ActuatorTest

This request is used to start the ActuatorTest.

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 31h |
| 1 | StartRoutineLocalIdentifier | 04h |
| 2 | ControlOption | 01h |
| 3 | ResultSelection | Xxh |
| 4..5 | ActuationValue 1 | Xxxxh |
| 6..7 | ActuatorNumber 1 | Xxxxh |
| 8..9 | ActuationValue 2 | Xxxxh |
| 10..11 | ActuatorNumber 2 | Xxxxh |
| 12..13 | ActuationTime (1LSB = 10ms) | Xxxxh |
| 14..15 | ActuationValue 3 | Xxxxh |
| 16..17 | ActuationNumber 3 | Xxxxh |
| 18..19 | ActuationValue 4 | Xxxxh |
| 20..21 | ActuatorNumber 4 | Xxxxh |

**Remark:** The pump motor relay must not be actuated for times shorter than 20ms. Shorter actuations could damage the motor relay due to very high current during the starting phase of the motor.

**Remark:** Bosch strongly recommends only using an ActuationValue of 0% and 100%.

**Remark:** A negative response has one of the following reasons: At least one valve or one driver doesn't exist or the control is not possible at this time because of safety reasons.

**Remark:** If the valve relay is out of order, the routine will be aborted.

**Example:**

- Start actuator test **without result** selection

| Request | Response | Comment |
|---|---|---|
| 31 04 01 **00** Xx Xx ... | 71 04 01 | -> Start actuation, Routine started by the tester |
| 31 04 03 | 71 04 03 | -> Status interrogation, control still in progress |
| . | . | |
| . | . | |
| 31 04 03 | 71 04 02 | -> Status interrogation, control complete, no result bytes sent |

- Start actuator test **with result** selection

| Request | Response | Comment |
|---|---|---|
| 31 04 01 **FF** Xx Xx ... | 71 04 01 | -> Start actuation, routine started by the tester |
| 31 04 03 | 71 04 03 | -> Status interrogation, control still in progress |
| . | . | |
| 31 04 03 | 71 04 02 Xx Xx ... | -> Status interrogation, control complete, result bytes sent |



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### 6.10.10 StartRoutineLocalIdentifier = DynamicTest

This request is used to start the Dynamic Test.

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 31h |
| 1 | StartRoutineLocalIdentifier | 05h |
| 2 | ControlOption | 01h |
| 3..4 | ActuationValue 1 | Xxxxh |
| 5..6 | ActuatorNumber 1 | Xxxxh |
| 7..8 | ActuationValue 2 | Xxxxh |
| 9..10 | ActuatorNumber 2 | Xxxxh |
| 11..12 | ActuationTime (1LSB = 10ms) | Xxxxh |
| 13..14 | ActuationValue 3 | Xxxxh |
| 15..16 | ActuationNumber 3 | Xxxxh |
| 17..18 | ActuationValue 4 | Xxxxh |
| 19..20 | ActuatorNumber 4 | Xxxxh |
| 21..22 | WaitTime (1LSB = 10ms) | Xxxxh |

### 6.10.11 StartRoutineLocalIdentifier = Test of the wheel speed sensors

This request is used to start the wheel speed sensor test.

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 31h |
| 1 | StartRoutineLocalIdentifier | 06h |
| 2 | ControlOption | 01h |
| 3 | Duration (1LSB = 100ms) | xxh |

#### 6.10.11.1 RequestRoutineResult

The Request Routine Results by Local Identifier gets the optional results of routines previously activated with the Start Routine.

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 31h |
| 1 | StartRoutineLocalIdentifier | xxh |
| 2 | ControlOption = Routine result | 03h |



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### 6.10.12 Positive Response

| PARAMETER: Control/RoutineStatus | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | Control/Routine Started successfully by the tester | 01h |
| 1 | Control/Routine completed | 02h |
| 1 | Control/Routine in progress | 03h |
| 1 | Control/Routine stopped | 04h |
| 1 | Control/Routine failure or Routine not run | 05h |

#### 6.10.12.1 Control Option = Start Routine and Stop Routine

For requests with Control option Start Routine (01h) and Stop Routine (02h), the positive response is returned in the following format.

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 71h |
| 1 | StartRoutineLocalIdentifier | xxh |
| 2 | ControlStatus<br>• Routine successfully Started 01h<br>• Routine in Progress 03h<br>• Routine Stopped 04h<br>• Routine Failure or Routine Not Run 05h | xxh |

**Remarks:**

- If the Tester send the request to Stop the Routine after routine is completed ,Positive Response(Control status =05h) will be returned
- If the Tester send the request to start the routine while another routine is in progress, Positive Response(Control status =05h) will be returned
- If the Tester send the request to start the routine while any temporary control in IOControlByLID service(SID 30h) is in progress, Positive Response(Control status =05h) will be returned.
- If the Tester send the request to start the routine while the same routine is in progress, Response(Control status =03h) will be returned.

#### 6.10.12.2 StartRoutineLocalIdentifier = GetSupportedLID

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 71h |
| 1 | StartRoutineLocalIdentifier | 00h |
| 2 | Data A describes the support of LID 01h ... 08h (MSB) | FCh |
| 3 | Data B describes the support of LID 09h ... 10h | 00h |
| 4 | Data C describes the support of LID 11h ... 18h | 00h |
| 5 | Data D describes the support of LID 19h ... 20h (LSB) | 00h |



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**Remark:** Supported PID (00h) gives the information about the supported local identifier from 01h – 20h (Bit 31 - PID 01h, Bit 30 – PID 02h … Bit 0 – PID 20h).

### 6.10.12.3 StartRoutineLocalIdentifier = Delayed Actuator, EvacAndFill and RepairBleeding

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 71h |
| 1 | StartRoutineLocalIdentifier (01h ,02h and 03h) | Xxh |
| 2 | ControlStatus | Xxh |

### 6.10.12.4 StartRoutineLocalIdentifier = Actuator Test, response without result

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 71h |
| 1 | StartRoutineLocalIdentifier | 04h |
| 2 | ControlStatus | Xxh |

### 6.10.12.5 StartRoutineLocalIdentifier = Actuator Test, response with result

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 71h |
| 1 | StartRoutineLocalIdentifier | 04h |
| 2 | ControlStatus | Xxh |
| 3..4 | Result, actuator 1 | Xxxxh |
| 5..6 | Result, actuator 2 | Xxxxh |
| 7..8 | Result, actuator 3 | Xxxxh |
| 9..10 | Result, actuator 4 | Xxxxh |

**Remark:** Result is a signal which informs about the former actuation of the corresponding actuator.
**Remark:** Bytes 3…10 are only sent if the response is “RoutineStatus = Routine completed”.

### 6.10.12.6 StartRoutineLocalIdentifier = Dynamic Test

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 71h |
| 1 | StartRoutineLocalIdentifier | 05h |
| 2 | ControlStatus | Xxh |
| 3..4 | Result, delta wheel speed front left (1LSB = (1/64)m/s) | Xxxxh |
| 5..6 | Result, delta wheel speed front right (1LSB = (1/64)m/s) | Xxxxh |
| 7..8 | Result, delta wheel speed rear left (1LSB = (1/64)m/s) | Xxxxh |
| 9..10 | Result, delta wheel speed rear right (1LSB = (1/64)m/s) | Xxxxh |

**Remark:** Result is a signed word information with a resolution of 1 LSB = 1/64 m/s = 0,05625 km/h and a range of -100 m/s … x … 100 m/s or -360 km/h … x … 360 km/h.
**Remark:** Bytes 3…10 are only sent if the response is “RoutineStatus = Routine completed”.



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### 6.10.12.7 StartRoutineLocalIdentifier = Test of the wheel speed sensors

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 71h |
| 1 | StartRoutineLocalIdentifier | 06h |
| 2 | RoutineStatus | xxh |
| 3..4 | Maximum speed of the front left wheel (1LSB = 0,01km/h) | xxxxh |
| 5..6 | Minimum speed of the front left wheel (1LSB = 0,01km/h) | xxxxh |
| 7..8 | Maximum speed of the front right wheel (1LSB = 0,01km/h) | xxxxh |
| 9..10 | Minimum speed of the front right wheel (1LSB = 0,01km/h) | xxxxh |
| 11..12 | Maximum speed of the rear left wheel (1LSB = 0,01km/h) | xxxxh |
| 13..14 | Minimum speed of the rear left wheel (1LSB = 0,01km/h) | xxxxh |
| 15..16 | Maximum speed of the rear right wheel (1LSB = 0,01km/h) | xxxxh |
| 17..18 | Minimum speed of the rear right wheel (1LSB = 0,01km/h) | xxxxh |

**Remark:** Bytes 3...18 are only sent if the response is “RoutineStatus = Routine completed”.

**Remark:** The test should be done during at least one complete wheel rotation.
**Remark:** If the test fails the minimum and maximum speed of a wheel speed sensor are set to 7FFFh.

**Remark:** For routines with an executing time exceeding $P2_{CAN\text{-}ECU}$ the ECU normally responds positive with RoutineStatus = Routine in progress within $P2_{CAN\text{-}ECU}$. During executing time the tester has to ask for the routine result by using the ControlOption = Request Routine Results. The ECU will either send a response with the RoutineStatus = Routine in progress or with the RoutineStatus = Routine completed and in addition if applicable with a result. The service request TesterPresent will as well be positively responded.

## 6.10.13 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | RequestServiceIdentifier | 31h |
| 2 | **ResponseCode**<br>SubFunctionNotSupported/InvalidFormat<br>ConditionsNotCorrect/RequestSeqError | <br>12h<br>22h |

Remarks

- At the end of each command, the elements which were tested have to be replaced in their OFF states.



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## 6.11 WriteDataByLocalIdentifier

### 6.11.1 Purpose

The service WriteDataByLocalIdentifier is used to write information into the ECU. The parameter RecordLocalIdentifier is used to select which information has to be written.

| PARAMETER: RecordLocalIdentifier | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | Filling-in status of the hydraulic block | 02h |
| 1 | Status of Test of end-of-line | 03h |

### 6.11.2 Request

#### 6.11.2.1 RecordLocalIdentifier = Filling-in status of the hydraulic block

This request is used to write a byte which indicates the filling-in status of the hydraulic block.

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 3Bh |
| 1 | RecordLocalIdentifier | 02h |
| 2 | Filling-in status:<br>• 00h : Filling-in not completed<br>• AAh : Filling-in completed and ok<br>• EEh : Filling-in completed and not ok | xxh |

**Remark:** Bosch delivery state is FFh.

#### 6.11.2.2 RecordLocalIdentifier = Status of Test of End-Of-Line

This request is used to write a byte which indicates the status if the end-of-line test.

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 3Bh |
| 1 | RecordLocalIdentifier | 03h |
| 2 | End-of-line test status:<br>• 00h : End-of-line test not completed<br>• AAh : End-of-line test completed and ok<br>• EEh : End-of-line test completed and not ok | xxh |

**Remark:** Bosch delivery state is FFh.



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### 6.11.3 Positive Response

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 7Bh |
| 1 | RecordLocalIdentifier | xxh |

### 6.11.4 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 3Bh |
| 2 | **Response Code**<br>General Reject<br>Sub Function Not Supported / Invalid Format<br>Conditions Not Correct / Request SequenceError<br>Response Pending<br>Security Access Denied | <br>10h<br>12h<br>22h<br>78h<br>33h |

**Remark:** If the reading of the EEPROM needs more than $P2_{CAN\text{-}ECU}$, the ECU will send a negative response 78h (RequestCorrectlyReceived-ResponsePending) until it has finished.



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## 6.12 TesterPresent

### 6.12.1 Purpose

The service TesterPresent is used to indicate the ECU that a tester is connected and diagnostic services and/or communication have to remain active. Sending the request keeps the ECU in a diagnostic session other than the StandardDiagMode.

| PARAMETER: ResponseRequired | | |
|---|---|---|
| **Size (Byte)** | **Description** | **Value** |
| 1 | ResponseRequired | 01h |
| 1 | NoResponseRequired | 00h |

### 6.12.2 Request

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 3Eh |
| 1 | ResponseRequired | 01h |

| TESTER → ECU: [SERVICE] – Request | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Identifier | 3Eh |
| 1 | NoResponseRequired | 00h |

### 6.12.3 Positive Response

| ECU → TESTER: [SERVICE] - Positive Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Service Positive Response Identifier | 7Eh |
| 1 | ResponseRequired | 01h |

### 6.12.4 Negative Response

| ECU → TESTER: [SERVICE] - Negative Response | | |
|---|---|---|
| **Byte** | **Contents** | **Value** |
| 0 | Negative Response Identifier | 7Fh |
| 1 | Request Service Identifier | 3Eh |
| 2 | SubFunctionNotSupported/InvalidFormat | 12h |



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# 7 List of DTCs

## 7.1 DTC List

| DTC | DTC (Hex) | Fault Symptoms | DTC Description |
|---|---|---|---|
| C1900 | 0x590004 | 0100 | ECU Voltage supply: high voltage |
| C1901 | 0x590104 | 0100 | ECU Voltage supply: low voltage |
| C1000 | 0x500004 | 0100 | ECU error (hardware, Micro-controller error) |
| C1010 | 0x501008 | 1000 | ECU error (general software problem) |
| C006B | 0x406B06 | 0110 | ABS implausible control (control time too long) |
| C0031 | 0x403108 | 1000 | Wheel-speed sensor, front left: (signal failure) out of range, lost, noise, Intermittent |
| C0032 | 0x403200 | 0000 | Wheel-speed sensor, front left: open |
| C00A0 | 0x40A000 | 0000 | Wheel-speed sensor, front left: short to GND |
| C00A1 | 0x40A100 | 0000 | Wheel-speed sensor, front left: short to UBATT |
| C00A9 | 0x40A900 | 0000 | Wheel-speed sensor, front left: undef (no exact failure reason detectable) |
| C0034 | 0x403408 | 1000 | Wheel-speed sensor, front right: (signal failure) out of range, lost, noise, Intermittent |
| C0035 | 0x403500 | 0000 | Wheel-speed sensor, front right: open |
| C00A2 | 0x40A200 | 0000 | Wheel-speed sensor, front right: short to GND |
| C00A3 | 0x40A300 | 0000 | Wheel-speed sensor, front right: short to UBATT |
| C00AA | 0x40AA00 | 0000 | Wheel-speed sensor, front right: undef (no exact failure reason detectable) |
| C0037 | 0x403708 | 1000 | Wheel-speed sensor, rear left: (signal failure)out of range, lost, noise, Intermittent |
| C0038 | 0x403800 | 0000 | Wheel-speed sensor, rear left: open |
| C00A4 | 0x40A400 | 0000 | Wheel-speed sensor, rear left: short to GND |
| C00A5 | 0x40A500 | 0000 | Wheel-speed sensor, rear left: short to UBATT |
| C00AB | 0x40AB00 | 0000 | Wheel-speed sensor, rear left: line failure (no exact failure reason detectable) |
| C003A | 0x403A08 | 1000 | Wheel-speed sensor, rear right: (signal failure)out of range, lost, noise, Intermittent |
| C003B | 0x403B00 | 0000 | Wheel-speed sensor, rear right: open |
| C00A6 | 0x40A600 | 0000 | Wheel-speed sensor, rear right: short to GND |
| C00A7 | 0x40A700 | 0000 | Wheel-speed sensor, rear right: short to UBATT |
| C00AC | 0x40AC00 | 0000 | Wheel-speed sensor, rear right: line failure (no exact failure reason detectable) |
| C1099 | 0x509904 | 0100 | Wheel-speed sensor generic error(sensor swap, multiple sensors error） |
| U0005 | 0xC00500 | 0000 | CAN Bus overvoltage |
| U0007 | 0xC00700 | 0000 | CAN Bus under voltage |
| C1001 | 0x500104 | 0100 | CAN hardware error |
| U0001 | 0xC00104 | 0100 | CAN Bus Off error |
| U1001 | 0xD10104 | 0100 | CAN Error Passive |
| C0010 | 0x401004 | 0100 | valve fault, inlet valve FL |



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| DTC | DTC (Hex) | Fault Symptoms | DTC Description |
|---|---|---|---|
| C0011 | 0x401104 | 0100 | valve fault, outlet valve FL |
| C0014 | 0x401404 | 0100 | valve fault, inlet valve FR |
| C0015 | 0x401504 | 0100 | valve fault, outlet valve FR |
| C0018 | 0x401804 | 0100 | valve fault, inlet valve RL |
| C0019 | 0x401904 | 0100 | valve fault, outlet valve RL |
| C001C | 0x401C04 | 0100 | valve fault, inlet valve RR |
| C001D | 0x401D04 | 0100 | valve fault, outlet valve RR |
| C1095 | 0x509504 | 0100 | Valve relay error |
| C0020 | 0x402004 | 0100 | Return pump fault |
| C0072 | 0x407208 | 1000 | General Valve fault (overheat protection, signal invalid, hardware fault ) |

NOTE:

Possible Fault Symptoms corresponds to

1) 0100 – No Signal
2) 1000 – Invalid Signal
3) 0000 – No fault symptom available for this DTC



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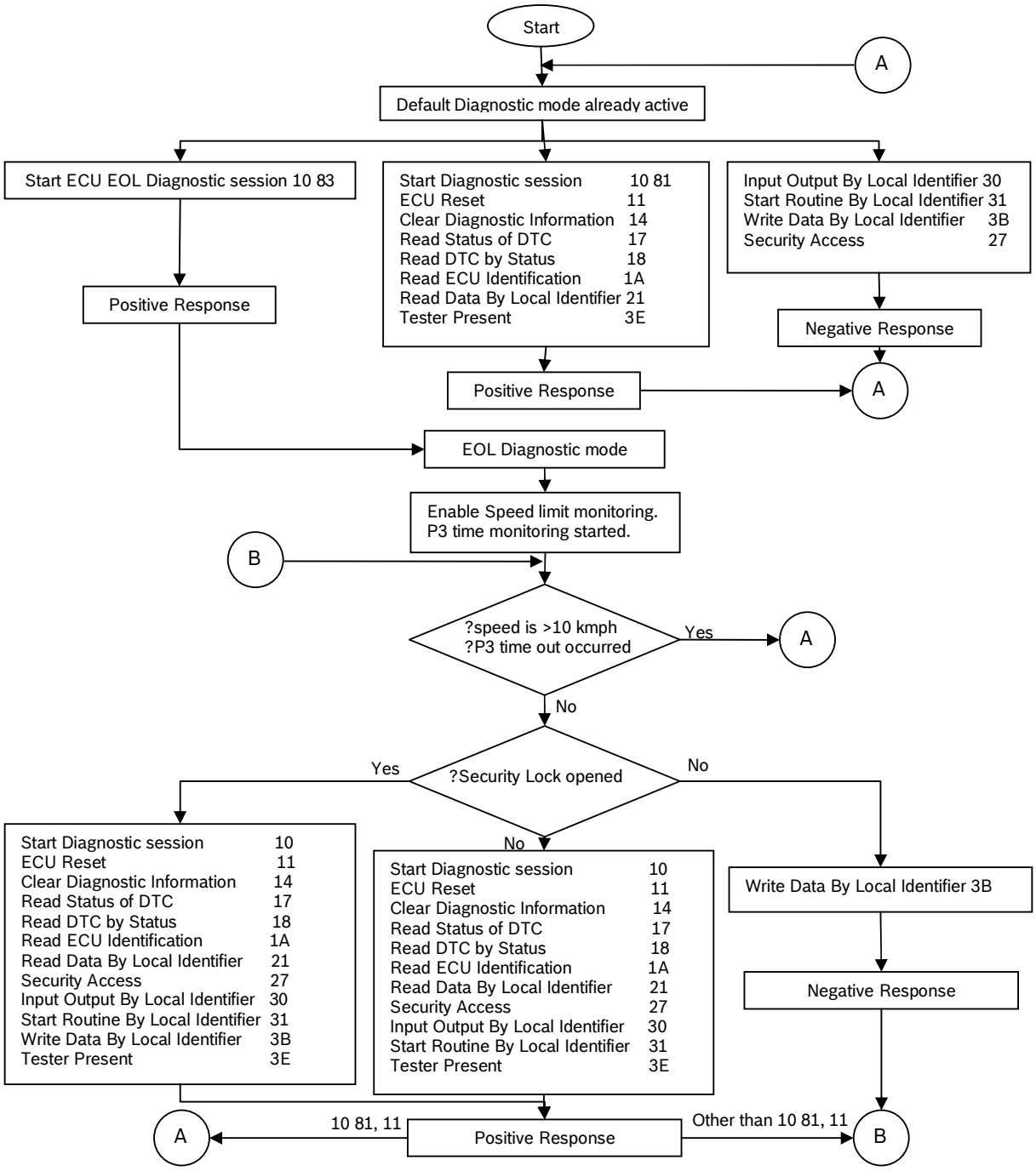
# 8 Diagnostic Flow Chart



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# 9 Appendix – A: Actuator Table

| Value | Description | Resolution/Comment |
|---|---|---|
| 0000h | No actuation* | 0x0000 |
| 0022h | Single control - Motor (Pump) | 0xFFFF – ON<br>0x0000 - OFF |
| 0030h | Single control - Front left valve (inlet) | 0xFFFF – ON<br>0x0000 – OFF |
| 0032h | Single control - Front left valve (outlet) | 0xFFFF – ON<br>0x0000 – OFF |
| 0034h | Single control - Front right valve (inlet) | 0xFFFF – ON<br>0x0000 – OFF |
| 0036h | Single control - Front right valve (outlet) | 0xFFFF – ON<br>0x0000 – OFF |
| 0038h | Single control - Rear right valve (inlet) | 0xFFFF – ON<br>0x0000 – OFF |
| 003Ah | Single control - Rear right valve (outlet) | 0xFFFF – ON<br>0x0000 - OFF |
| 003Ch | Single control - Rear left valve (inlet) | 0xFFFF – ON<br>0x0000 – OFF |
| 003Eh | Single control - Rear left valve (outlet) | 0xFFFF – ON<br>0x0000 – OFF |
| 0002h | Single control – ABS Safety Lamp | 0xFFFF – ON<br>0x0000 – OFF |
| 000Eh | Single control – EBD Safety Lamp | 0xFFFF – ON<br>0x0000 - OFF |

**Remark:** * Use if less than four actuators are to be actuated by an actuator test or dynamic test request.



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